B. Prefiquence

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: greedy, two pointers | https://codeforces.com/problemset/problem/1968/B

You are given two binary strings *a* and *b*. A binary string is a string consisting of the characters '0' and '1'.

Your task is to determine the maximum possible number *k* such that a prefix of string *a* of length *k* is a subsequence of string *b*.

A sequence *a* is a subsequence of a sequence *b* if *a* can be obtained from *b* by the deletion of several (possibly, zero or all) elements.

Input

The first line consists of a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The first line of each test case contains two integers n and m ($1 \le n, m \le 2 \cdot 10^5$) — the length of string *a* and the length of string *b*, respectively.

The second line of each test case contains a binary string *a* of length n.

The third line of each test case contains a binary string *b* of length m.

It is guaranteed that the sum of values n over all test cases does not exceed $2 \cdot 10^5$. Similarly, the sum of values m over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output a single number — the maximum k, such that the first k characters of *a* form a subsequence of *b*.

Example

Input
6 5 4 10011 1110 3 3 100 110 1 3 1 111 4 4 1011 1111 3 5 100 11010 3 1 100 0

Output
2 2 1 1 3 0

Note

In the first example, the string '10' is a subsequence of '1color*red*11color*red*0' but the string '100' is not. So the answer is 2.

In the fifth example, *a*='100', *b*='1color*red*101color*red*0', whole string *a* is a subsequence of string *b*. So the answer is 3.

In the sixth example, string *b* does not contain '1' so the answer is 0.